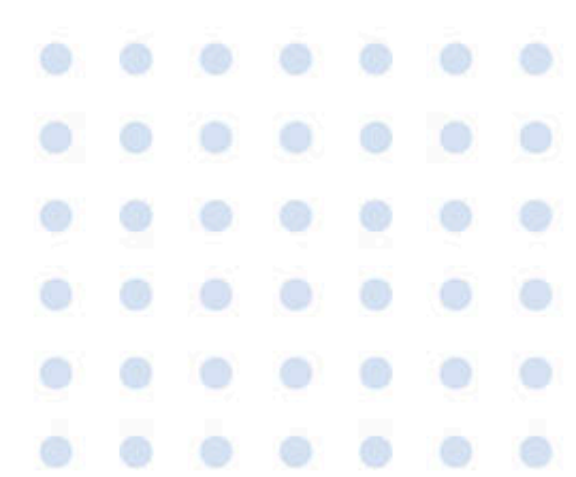




#CPT1 

Basic drug interaction & clinical approach

Apirat Muangkasem, BCP





Recall



Describe/Understand

- Definition and mechanism
- Principles of information search
- Assessment of sign and symptoms, laboratory findings, plan, and monitoring from scenarios



of **drug interaction**

Learning objectives

Importance

Article

Full-text available

Fluconazole-Warfarin Interaction: A case report with deadly consequences

June 2017 · Australasian Medical Journal 10(6)

DOI: 10.21767/AMJ.2017.3049

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Elliot V Hersh

- "The patient was confused and driving his car erratically. His son rushed him to the hospital where a CAT scan revealed **a large frontal intraparenchymal hemorrhage.**"
- "His **INR** was a **9.6**"

Importance

Case Report

Drug-Drug Interaction of Ergotamine with a Combination of Darunavir, Abacavir, and Lamivudine Causing a Fatal Vasospastic Ischemia

Elisabeth Petit ¹, Karen Schoonheydt,² Philippe Meert,¹ and Marie Van Laer¹

- The patient developed **severe limb ischemia, cerebral ischemia, and metabolic encephalopathy**.
- Due to evolving **cerebral edema** as a result of the ischemia, the patient **developed brain herniation and died** shortly after.

Definition

- **A drug interaction** occurs when a patient's response to a drug is modified by **food, nutritional supplements, formulation excipients, environmental factors, other drugs** or **disease**.
- Interactions between drugs may be **beneficial** or **harmful**. Harmful drug-drug interactions are important as they cause 10–20% of the adverse drug reactions requiring hospitalization and **they can be avoided**.

Mechanism

Pharmacokinetic interaction:

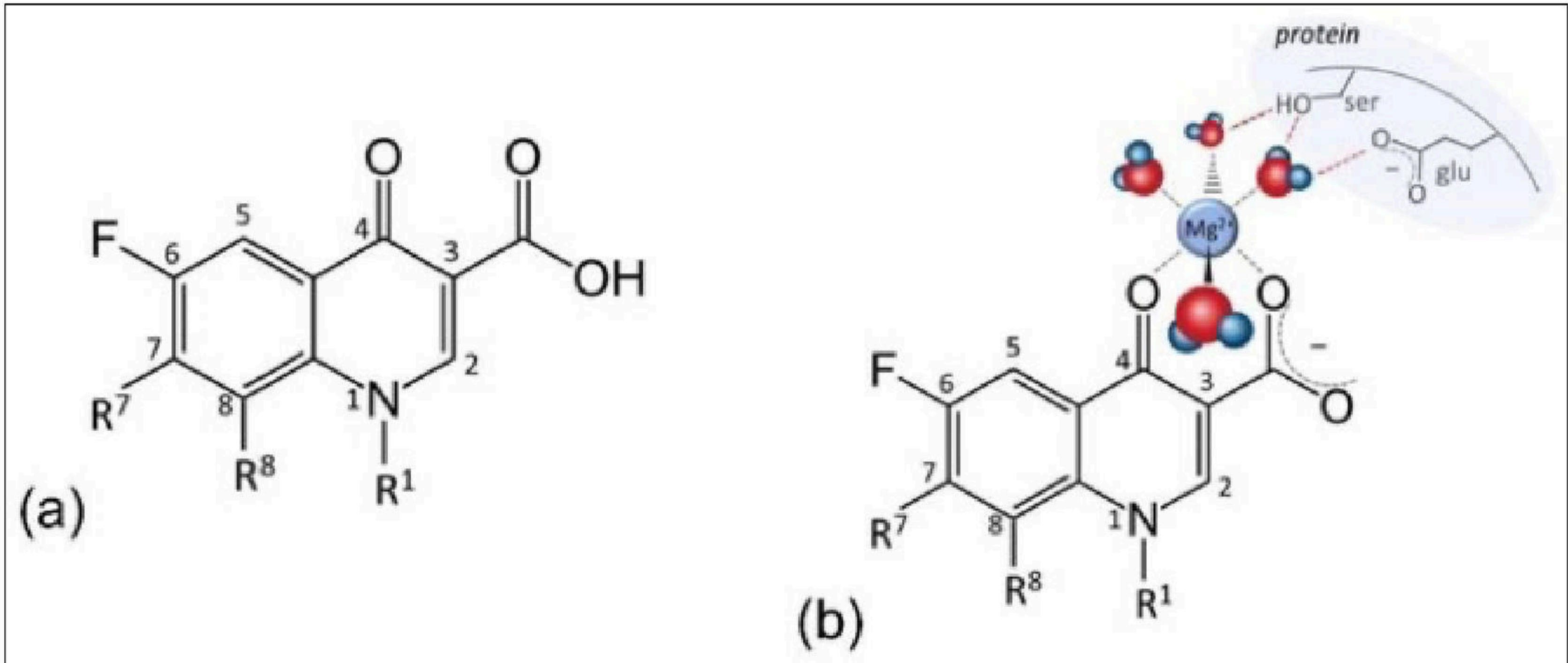
- Influencing of **absorption, distribution, metabolism, and elimination** of the drug

Pharmacodynamic interaction:

- Influencing each **drug effect directly**

Mechanism

Pharmacokinetic interaction: Influencing of **absorption** → complex formation



Mechanism

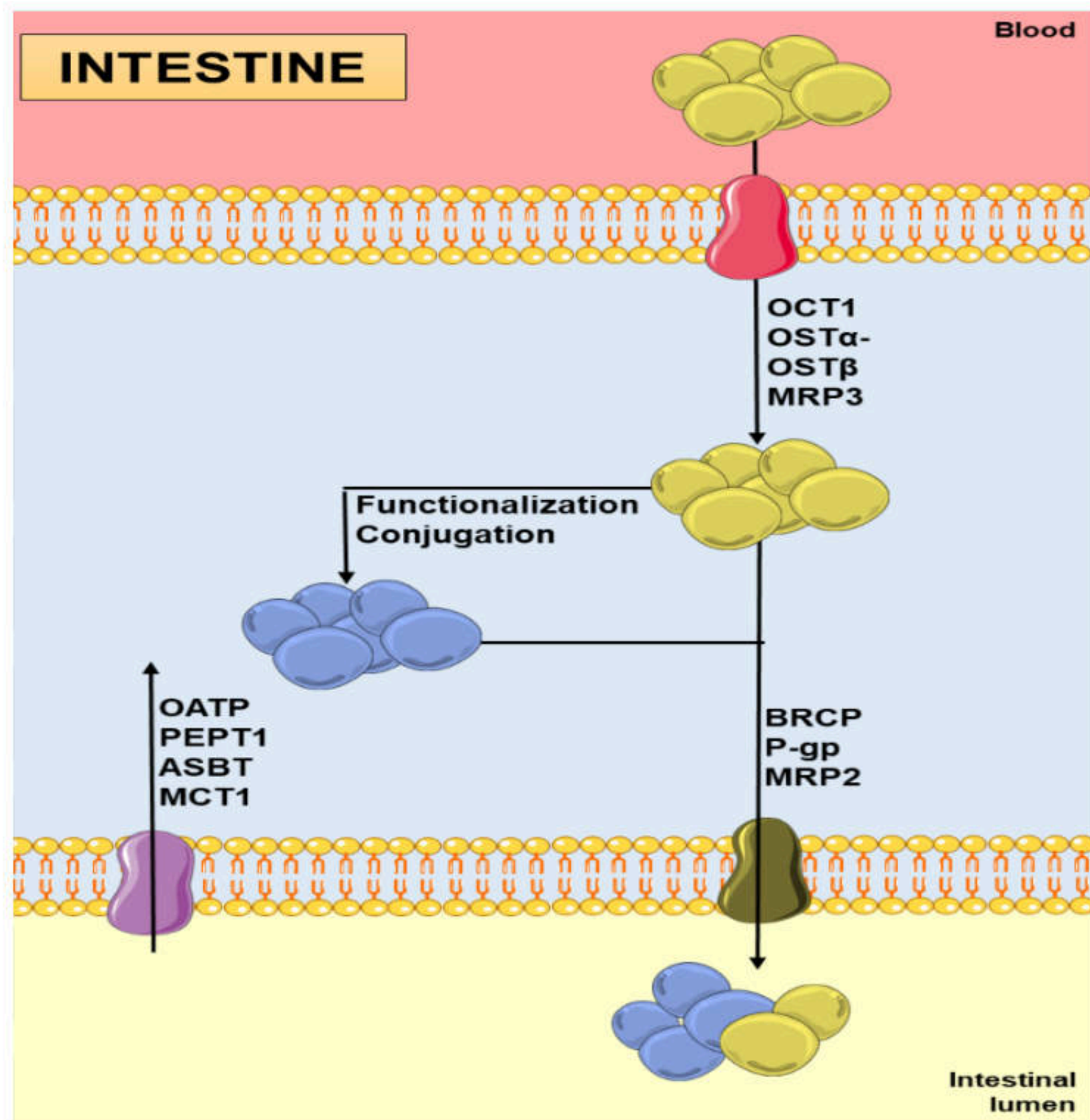
Pharmacokinetic interaction: Influencing of **absorption** → transporters in the intestine

- **Example of transporters:**

- P-glycoprotein (**P-gp**)
- Organic anion transporting polypeptide (**OATP**)
- Peptide transporter (**PEPT**)
- Breast cancer resistance protein transporter (**BCRP**)
- Multidrug resistance-associated protein (**MRCP**)

Mechanism

Pharmacokinetic interaction: Influencing of **absorption** → transporters in the intestine



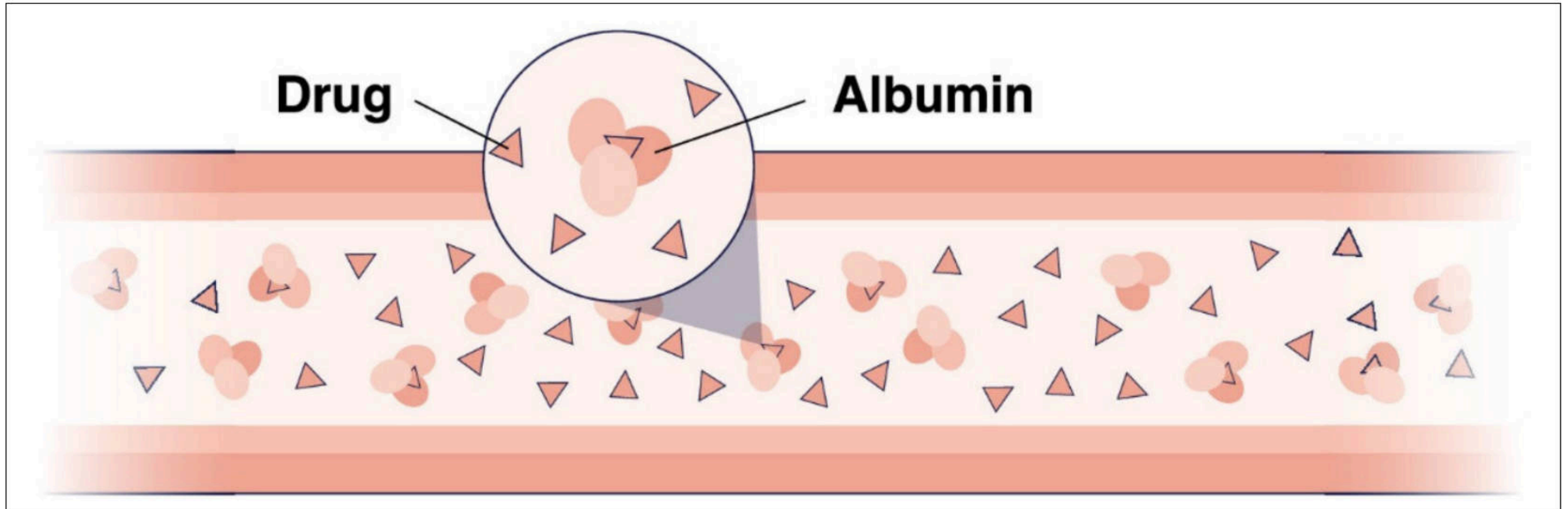
Mechanism

Pharmacokinetic interaction: Influencing of **absorption** → transporters in the intestine

P-gp substrates	P-gp inducers	P-gp inhibitors
Dabigatran	Carbamazepine	Amiodarone
Digoxin	Phenobarbital	Ketoconazole
Tacrolimus	Phenytoin	Itraconazole
	Rifampicin	Erythromycin
	Efavirenz	Clarithromycin
		Ritonavir

Mechanism

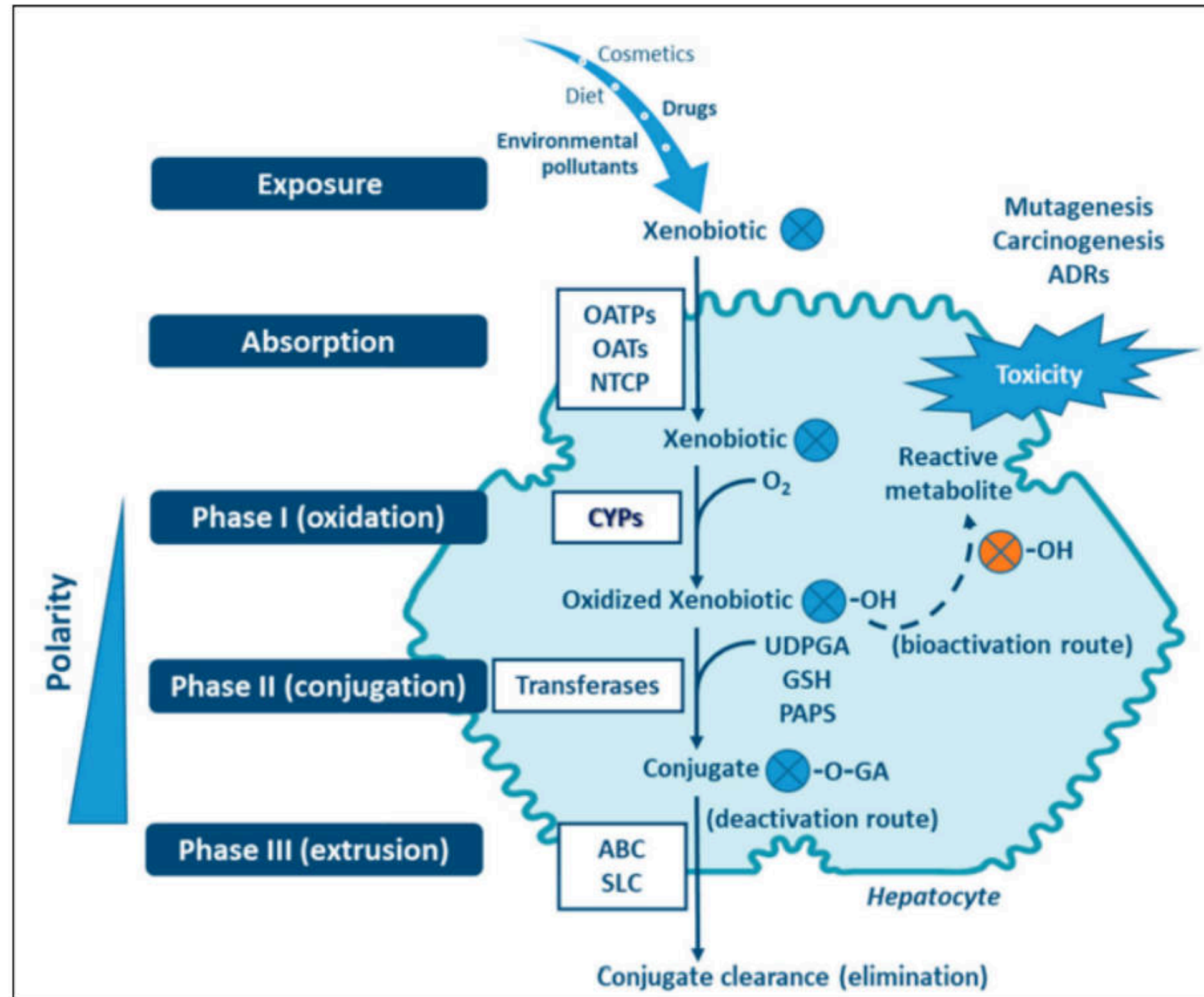
Pharmacokinetic interaction: Influencing of **distribution** → protein binding



Mechanism

Pharmacokinetic interaction: Influencing of **metabolism** → Cytochrome P450 enzyme (CYP450)

- **CYP substrates**



Mechanism

Pharmacokinetic interaction: Influencing of **metabolism** → Cytochrome P450 enzyme (CYP450)

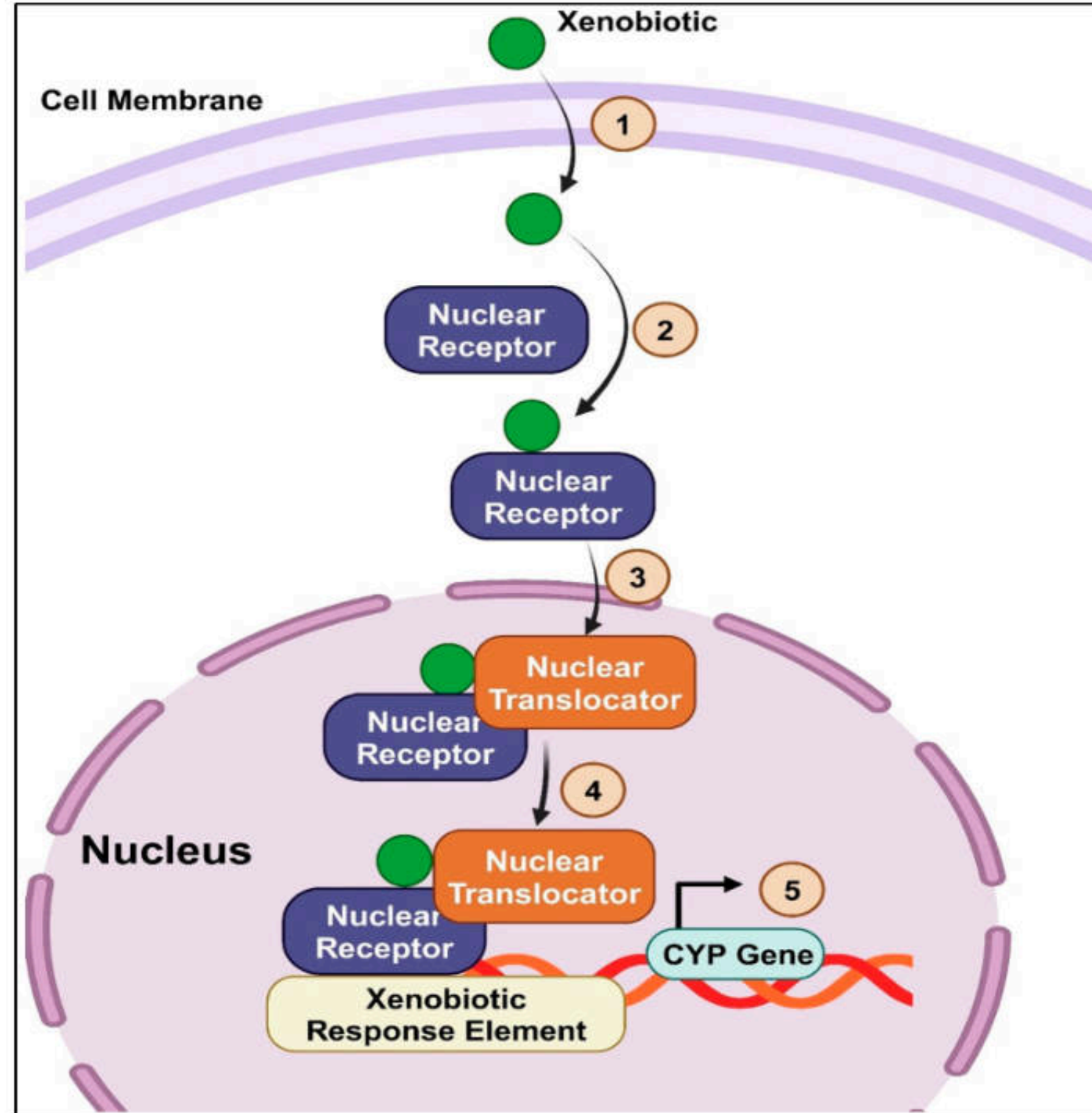
- CYP substrates**

CYP1A2	CYP2C9	CYP2C19	CYP2D6	CYP3A4/5
Clozapine	Ibuprofen	Omeprazole	Metoprolol	Clarithromycin
Imipramine	Diclofenac	Lansoprazole	Amitriptyline	Diazepam
Naproxen	Naproxen	Amitriptyline	Duloxetine	Amlodipine
Theophylline	Celecoxib	Diazepam	Paroxetine	Diltiazem
	Glipizide	Clopidogrel	Venlafaxine	Cyclosporine
	Losartan	Cyclophosphamide	Haloperidol	Tacrolimus
	Cyclophosphamide	Phenytoin	Risperidone	Ritonavir
	Phenytoin		Codeine	Simvastatin
	Sulfamethoxazole		Tramadol	Apixaban
	Warfarin		Tamoxifen	Sildenafil

Mechanism

Pharmacokinetic interaction: Influencing of **metabolism** → Cytochrome P450 enzyme (CYP450)

- **CYP inducers**



Mechanism

Pharmacokinetic interaction: Influencing of **metabolism** → Cytochrome P450 enzyme (CYP450)

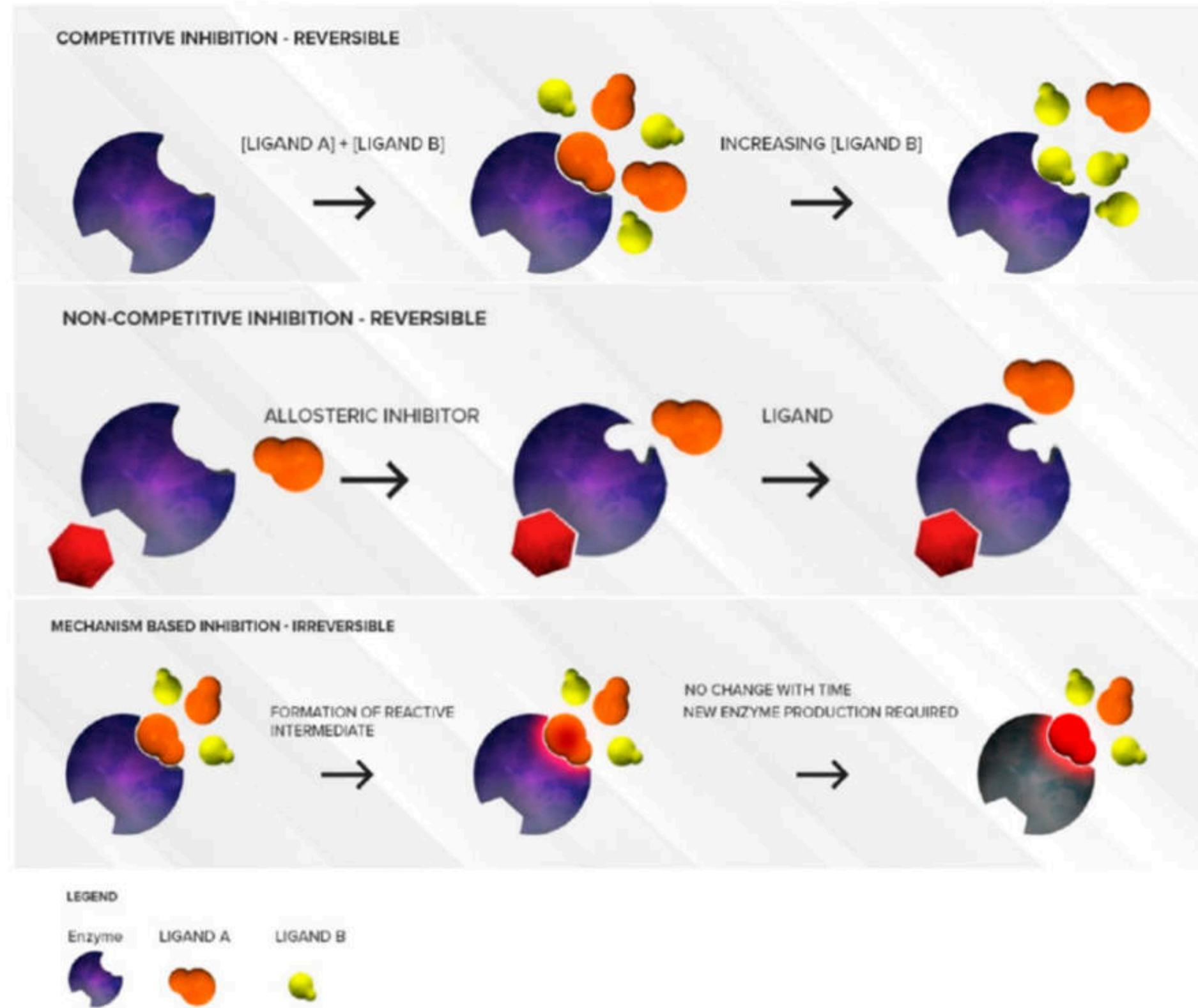
- **CYP inducers**

CYP1A2	CYP2C9	CYP3A4/5
Tobacco	Rifampicin	Carbamazepine
Omeprazole		Efavirenz
		St. John's wort
		Phenobarbital
		Phenytoin
		Rifampicin

Mechanism

Pharmacokinetic interaction: Influencing of **metabolism** → Cytochrome P450 enzyme (CYP450)

- **CYP inhibitors**



Mechanism

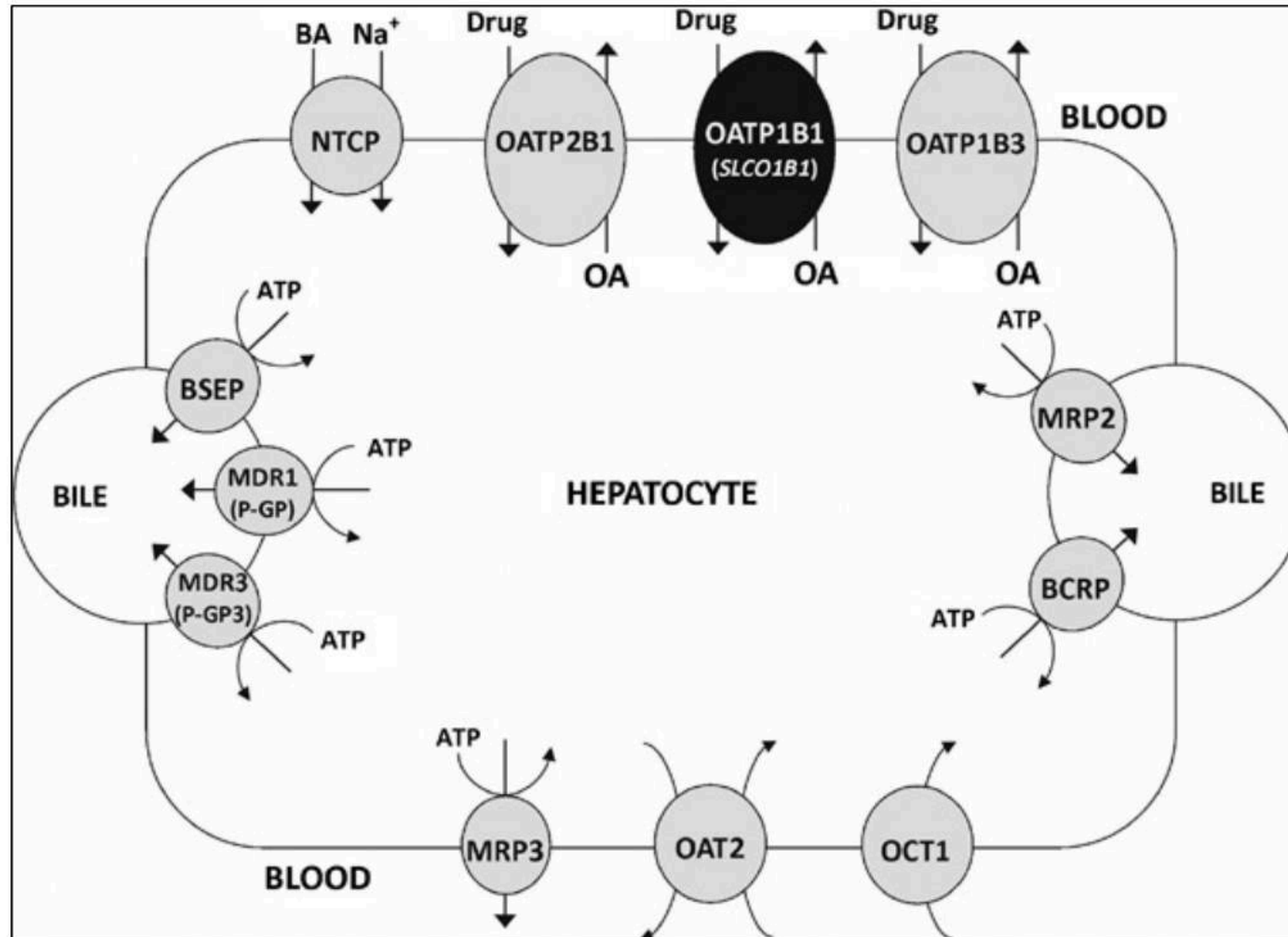
Pharmacokinetic interaction: Influencing of **metabolism** → Cytochrome P450 enzyme (CYP450)

- CYP inhibitors**

CYP1A2	CYP2C9	CYP2C19	CYP2D6	CYP3A4/5
Ciprofloxacin	Amiodarone	Fluoxetine	Duloxetine	Ritonavir
Amiodarone	Fluconazole	Omeprazole	Paroxetine	Clarithromycin
Cimetidine	Isoniazid	Lansoprazole	Amiodarone	Ketoconazole
Fluvoxamine		Voriconazole	Quinidine	Fluconazole
			Ritonavir	Itraconazole
				Voriconazole
				Amiodarone
				Cimetidine
				Diltiazem
				Verapamil

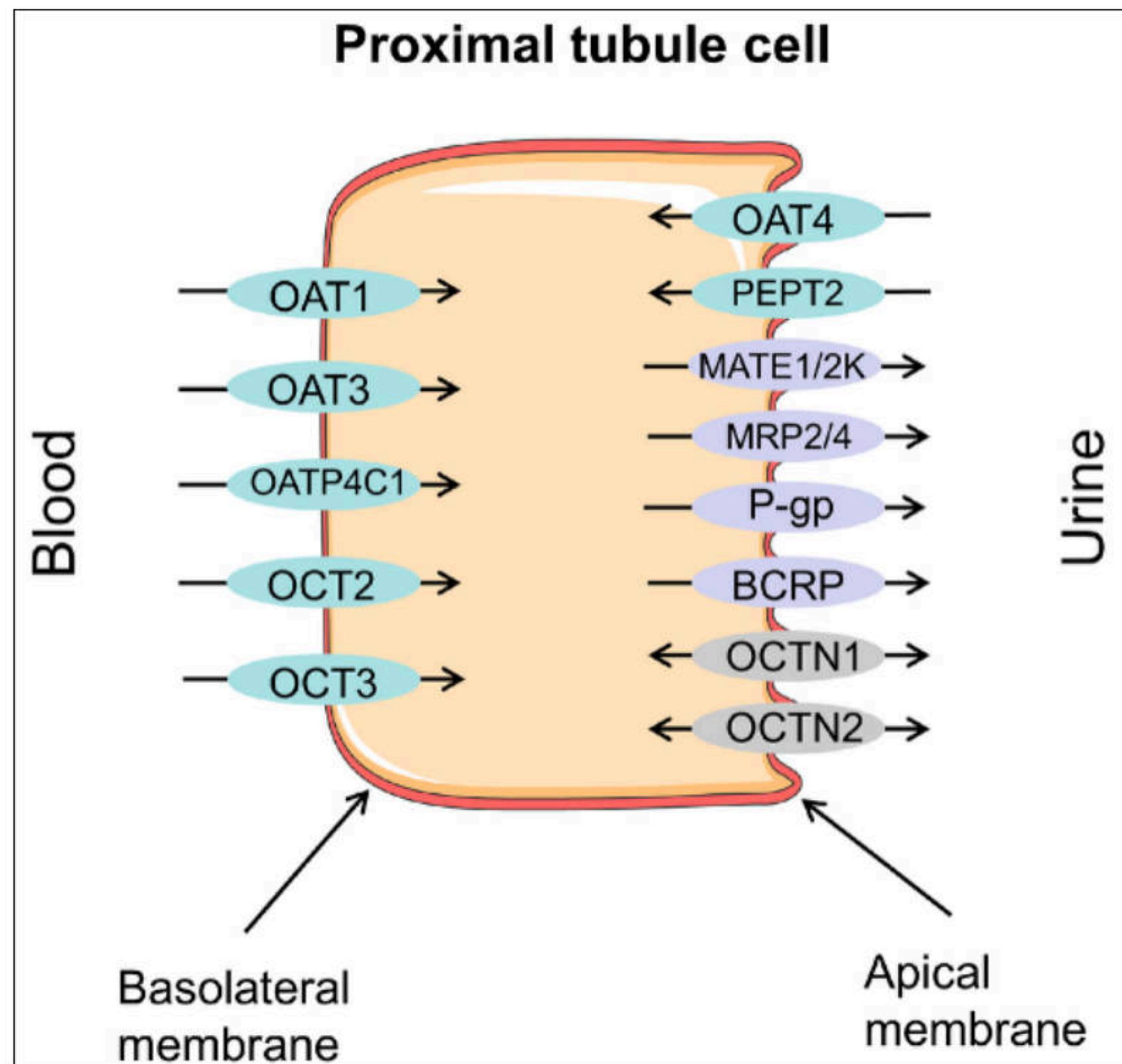
Mechanism

Pharmacokinetic interaction: Influencing of **metabolism** → Transporters in the liver



Mechanism

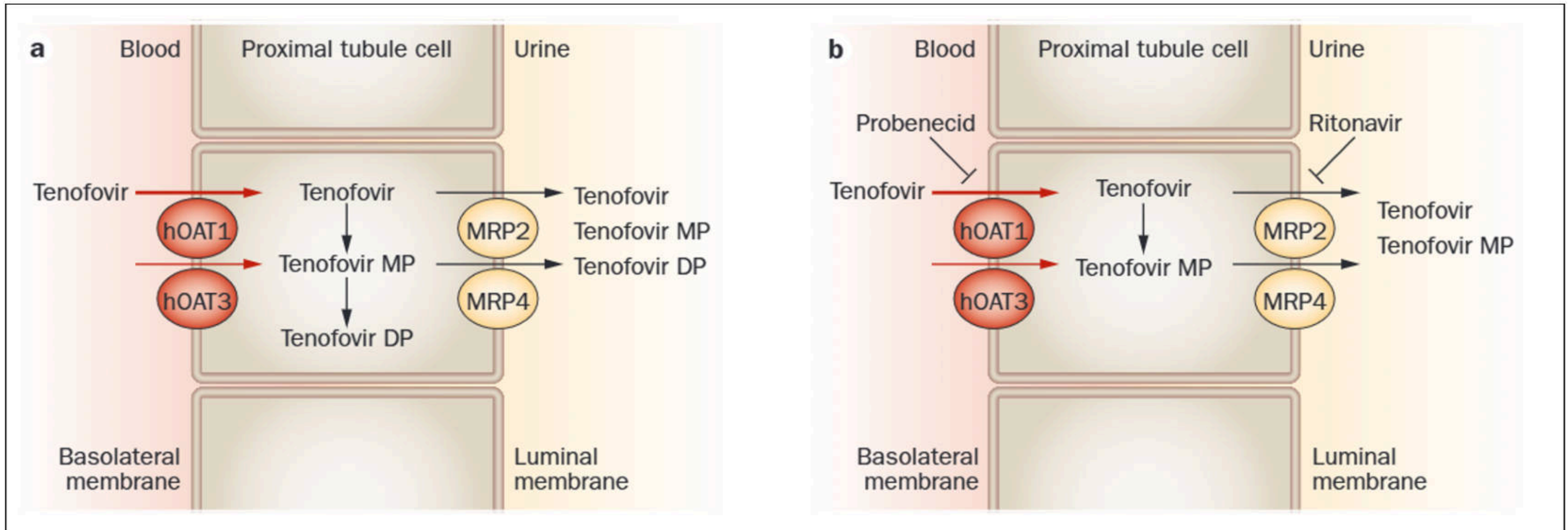
Pharmacokinetic interaction: Influencing of **elimination** → Transporters in the kidney



Mechanism

Pharmacokinetic interaction: Influencing of **elimination** → Transporters in the kidney

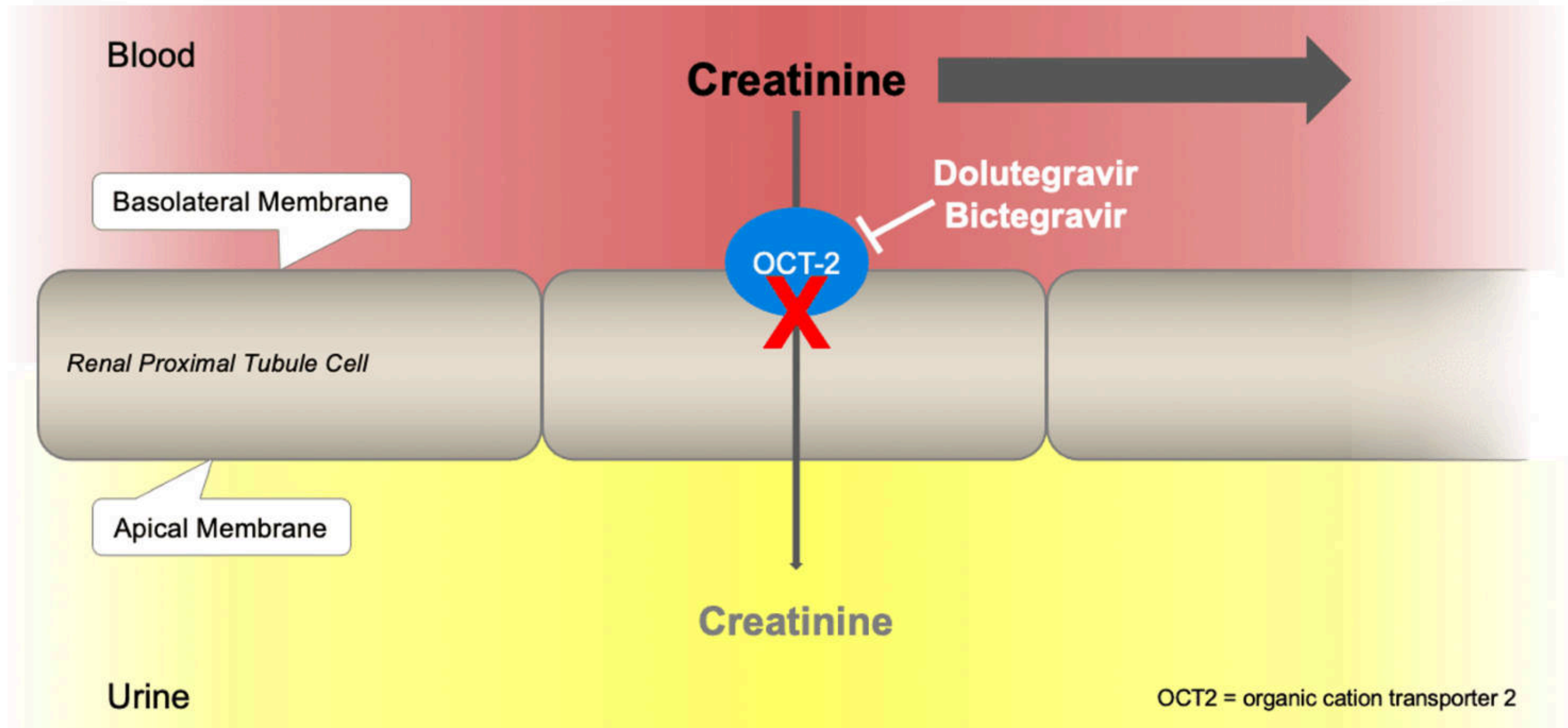
- Organic **anion** transporting polypeptide (**OATP**)



Mechanism

Pharmacokinetic interaction: Influencing of **elimination** → Transporters in the kidney

- Organic **cation** transporting polypeptide (**OCTP**)



Mechanism

Pharmacodynamic interaction:

- **Additive** interaction
- **Antagonistic** interaction
- **Synergistic** interaction

Mechanism

Pharmacodynamic interaction: **Additive** interaction

Additive interactions		
Substrate I	Substrate II	Effects
NSAIDs	Warfarin	Increased risk of bleeding
NSAIDs	Glucocorticoids	Increased risk of gastric bleeding
ACE inhibitors	Spironolactone	Hyperkalemia
SSRIs	Triptans	Serotonin syndrome
Quinolones	Macrolides	QT-interval prolongation

Mechanism

Pharmacodynamic interaction: Antagonistic interaction

Antagonistic interactions		
Substrate I	Substrate II	Effects
Aspirin	Ibuprofen	Reduced antiplatelet effect
ACE inhibitors	NSAIDs	Reduced blood pressure lowering effect
Warfarin	Vitamin K	Reduced anticoagulant effect

Clinical approach

Avoid undesired drug interactions

- Ensure you have a **full drug history** including over the counter (**OTC**), **vitamins**, **food**, and **herbal products**
- Drugs with a **narrow therapeutic index (TI)** are particularly venerable
- Knowledge of the **pharmacological effects** are crucial for pharmacodynamic interaction

Clinical approach

Resources: Textbook



BOOK

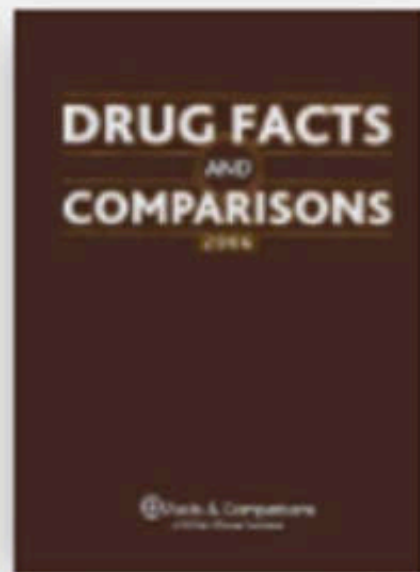
Drug interaction facts : the authority on drug interactions

Tatro, David S., ed.

c2013

 [Available at PY-Pharmacy Library Reference - Internal Use Only \(QV38 D794 2013\)](#) >

 [Book reviews \(1\)](#) >



BOOK

Drug : facts and comparisons

Facts and Comparisons Division

1981-<2017>

 COURSE

 [Available at PY-Pharmacy Library General Stacks \(QV772 D794k 1992\) and other locations](#) >

Clinical approach

Resources: Micromedex

Merative **Micromedex**[®]

My Subscription | Gateway | Training Center | Help | Mobile Application Access | Logout

HomeDrug InteractionsIV CompatibilityDrug IDDrug ComparisonNeoFax[®] / PediatricsTox & Drug Product LookupCalculatorsFormulary

Search Drug, Disease, Toxicology, and more

AllDrugDiseaseToxicology

Micromedex Assistant
Search Micromedex drug information

Type a quick question...→

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 **Latest News**

- New/Expanded Indications...
- New Drug Approvals August/September
- Your DID YOU KNOW for August
- Patient Education (CareNotes) Content...
- Content Update Highlights

Read Top News▶

 **Support & Training**

- Training Center
- Micromedex Compendia Resources
- Citing Micromedex

Support Request▶

 **Resources**

- Do Not Crush Drugs List
- Extravasation
- Drug Classes
- Drug Consults
- P&T Quik Reports
- Boxed Warnings
- Dosage and Class Comparison Tables
- REMS

Download Mobile Apps

Feedback

27

Clinical approach

Resources: Micromedex

Drug Interaction Results

← Modify Interactions

Print

Refine by:

Drugs: All

Severity: All

Documentation: All

Type: All

Jump To:

DRUG-DRUG (1)

|

Ingredient Duplication (0)

|

ALLERGY (0)

|

FOOD (14)

|

ETHANOL (1)

|

LAB (0)

|

TOBACCO (1)

|

PREGNANCY (2)

|

LACTATION (2)

Drug-Drug Interactions (1)

Drugs:	Severity:	Documentation:	Summary:
FLUCONAZOLE – WARFARIN SODIUM	S Major	Excellent	Concurrent use of FLUCONAZOLE and WARFARIN may result in increased INR and increased risk of bleeding events.

Ingredient Duplication (None found)

Drug-ALLERGY Interactions (None found)

Drug-FOOD Interactions (14)

Drugs:	Severity:	Documentation:	Summary:
WARFARIN SODIUM	S Major	Excellent	Concurrent use of WARFARIN and VITAMIN K FOODS may result in altered anticoagulant effectiveness.
WARFARIN SODIUM	S Major	Fair	Concurrent use of WARFARIN and GRAPEFRUIT JUICE may result in increased warfarin exposure and an increased INR.
WARFARIN SODIUM	S Major	Good	Concurrent use of WARFARIN and CRANBERRY JUICE may result in an increased risk of bleeding.

Feedback

Feedback

Clinical approach

Resources: UpToDate

UpToDate®

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Practice Changing Updates

View All

ONCOLOGY: [Aspirin for stage I to III PIK3CA-mutated colorectal cancer](#) (August 2025)

INFECTIOUS DISEASES: [Updated guidelines for prophylaxis after a nonoccupational exposure to HIV](#) (July 2025)

What's New

View All

Drug therapy

Drug Interactions

Patient Education

Calculators

Clinical approach

Resources: UptoDate

< Back

Drug Interactions

Print

Item(s)

Q Enter Item Name

Add

X Warfarin

X Fluconazole

Clear

Analyze

Display complete list of interactions for an individual item by clicking item name.

X

Avoid combination

D

Consider therapy modification

C

Monitor therapy

B

No action needed

A

No known interaction

More about Risk Ratings

▼

1 Result

View interaction detail by clicking on link(s) below.

D

Warfarin (Vitamin K Antagonists)

Fluconazole

DISCLAIMER: Readers are advised that decisions regarding drug therapy must be based on the individual patient's clinical situation, current product information), and changing medical practices.

Filter Results by Item ▼

X

Avoid Combination
Data demonstrate that the specified agents may interact with each other in a clinically significant manner. The risks associated with concomitant use of these agents usually outweigh the benefits. Concurrent use of these agents should generally be avoided.

D

Consider Therapy Modification
Data demonstrate that the two medications may interact with each other in a clinically significant manner. A patient-specific assessment must be conducted to determine whether the benefits of concomitant therapy outweigh the risks. Specific actions must be taken in order to realize the benefits and/or minimize the risks resulting from concomitant use of the agents. These actions may include aggressive monitoring, empiric dosage changes, or choosing alternative agents.

C

Monitor Therapy
Data demonstrate that the specified agents may interact with each other in a clinically significant manner. The benefits of concomitant use of these two medications often outweigh the risks. An appropriate monitoring plan should be implemented to identify potential negative effects. Dosage adjustments of one or both agents may be needed in some patients.

B

No Action Needed
Data demonstrate that the specified agents may interact with each other, but there is little to no evidence of clinical concern resulting from their concomitant use.

A

No Known Interaction
Data have not demonstrated either pharmacodynamic or pharmacokinetic interactions between the specified agents

30

Clinical approach

Resources: UptoDate

Title Vitamin K Antagonists / Fluconazole

Risk Rating D: Consider therapy modification

Summary Fluconazole may increase serum concentrations of Vitamin K Antagonists. **Severity** Major **Reliability Rating** Highest

Patient Management Consider alternatives to fluconazole in patients receiving therapy with vitamin K antagonists that are less likely to influence vitamin K antagonists activity (itraconazole, ketoconazole, and posaconazole may be less likely to interact). If coadministered, consider an empiric reduction in vitamin K antagonist dose of 10% to 20% together with increased monitoring of anticoagulant response (ie, INR testing, signs/symptoms of bleeding) to guide any further dose adjustment. One simulation study has suggested that larger dose reductions may be necessary, but that gradual dose reductions are superior to an immediate decrease to the predicted dose requirement. Increased monitoring for increased therapeutic/toxic effects of vitamin K antagonists is critical whenever using fluconazole in combination.

Vitamin K Antagonists Interacting Members Acenocoumarol, Phenindione, Phenprocoumon, Warfarin

Discussion Combined use of fluconazole with warfarin was associated with approximately a 2-fold increased odds of hospitalization for gastrointestinal bleeding (compared to warfarin plus cephalexin) in a case-control study of Medicaid data for over 300,000 warfarin users.¹ The time period of greatest risk for hospitalization was 11 to 15 days after fluconazole initiation. In a pharmacoepidemiologic study involving 38,762 continuous users of warfarin 65 years or older, those exposed to azole antifungals (n=17) within the previous 15 days had an increased risk of bleeding requiring hospitalization (OR = 4.57) compared to matched controls (n=8), although the specific azole antifungal agents used were not identified.²

In a pharmacokinetic study of 6 healthy volunteers, fluconazole (400 mg daily) coadministered with warfarin (0.75 mg/kg single dose) increased the S-warfarin and R-warfarin AUC 2.8-fold and 2.1-fold, respectively.³ Studies in patients receiving warfarin therapy found that low-dose fluconazole (100 mg to 150 mg) increased the average prothrombin time (PT) 34% to 39%, but several patients experienced an INR greater than 4, bleeding episodes, or a need to modify warfarin therapy or discontinue fluconazole.^{4,5} The area under the prothrombin time curve (PT AUC) (as a measure of clinical warfarin effect), over 168 hours, was increased 10% to 100% in 13 healthy volunteers receiving a single dose of warfarin (15 mg) following a 7-day course of fluconazole (200 mg daily).⁶ The mean INR increased by 0.83 after initiation of fluconazole in 413 patients treated with warfarin and who initiated fluconazole for the treatment of oral candidiasis.⁷ Initiation of fluconazole also increased the proportion of patients with an INR > 5 by 3.6-fold. In a retrospective study of warfarin users, use of CYP2C9 inhibitors or inducers (including fluconazole) was associated with more variable INRs and less time in the therapeutic range (TTR) in normal CYP2C9 metabolizers (*1/*1).⁸ There was no association between INR variability or TTR in patients taking CYP2C9 inhibitors or inducers in patients with any CYP2C9 loss of function allele (*1/*2, *1/*3, or *2/*2). Numerous cases of bleeding due to this presumed interaction have also been reported.^{9,10,11,12,13}

Regarding optimal clinical management of this interaction, a series of simulation studies predicted that a gradual reduction in warfarin dose coincident with the start of fluconazole therapy would result in a more stable anticoagulant effect than a one-time reduction in warfarin dose.¹⁴ Modeling the initiation of fluconazole 400 mg daily, the simulation showed that administration of 80% of warfarin dose on day 1 of the combination, followed then by daily warfarin doses corresponding to 50%, 43%, and 36% of the pre-fluconazole dose, and then staying at 36% of the pre-fluconazole dose, was superior (at maintaining baseline S-warfarin concentrations) than a one-time reduction in warfarin dose to 36% of its pre-fluconazole levels.¹⁴ Simulations for other fluconazole doses (50 mg to 400 mg per day) predicted that warfarin dose reductions of 21% to 67% could be required for longer-term combination therapy.

The mechanism of this interaction is likely due to fluconazole-mediated inhibition of CYP2C9 and CYP3A4, enzymes primarily responsible for S-warfarin and R-warfarin metabolism, respectively.¹⁵

Clinical approach

Assessment & Plan

- **Avoid/terminate use**
- **Modify drugs**
 - Alternative drug
 - Dose increase/decrease
 - Time interval
- **Monitoring**
 - Sign and symptoms
 - Laboratory findings
 - Time to monitor (onset/duration)

Clinical approach

Alarming drug list: check DDIs

- **Antacids, calcium, ferrous** → complex formation
 - Fluoroquinolones, tetracyclines, levothyroxine
- **PPIs** → altered stomach pH → altered some drug absorption
 - Itraconazole
- **Warfarin** → narrow TI / high protein binding / CYP450 substrate
- **Digoxin** → drug affect potassium level
 - Loop diuretics, thiazides
- **CYP450 inducers:**
 - Rifampin, phenytoin, carbamazepine, phenobarbital, St. John's Wort
- **CYP450 inhibitors:**
 - Amiodarone, macrolides, azole antifungals, ritonavir, grapefruit juice

Case study



A 52-year-old male complains of nose bleeding and bruises on his arms.

- **PMH:** Atrial fibrillation, Hypertension
- **MH:**
 - Warfarin 5 mg 1 tablet once daily
 - Enalapril 5 mg 1 tablet once daily
 - Metronidazole 500 mg 1 tablet 3 times a day (started 5 days ago by his dentist)
- **Lab:** Platelet $210,000 / \text{mm}^3$, INR 8

Answer these questions:

- DDI mechanism:
- Risk rating/severity:
- What is your management:

Case study



A 32-year-old male returns to follow up with HIV clinic

- **PMH:**
 - HIV infection (started treatment 5 years ago)
 - Tuberculosis lung infection (started treatment 2 weeks ago)
- **MH:**
 - TDF/3TC/DTG 300/300/50 mg 1 tablet once daily
 - Isoniazid 100 mg 3 tablets once daily
 - Rifampicin 450 mg 1 tablet once daily
 - Pyrazinamide 500 mg 2 tablets once daily
 - Ethambutol 400 2 tablets once daily
 - Vitamin B6 50 mg 1 tablet once daily
- **Lab:** HIV viral load 12,500 copies/mL (last visited: undetected), CD4 500 cells/mm³

Case study



Answer these questions:

- DDI mechanism:
- Risk rating/severity:
- What is your management:



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